

Evaluation of Multiple Optimized AIGC Approaches in Low-Altitude Drone-based Riparian Waste Monitoring

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Recently, LMM (large multimodal model, which can “understand” image, text or other types of modals) has been applied in camera-based riparian environmental monitoring (4K camera or drones). Concurrently, LMM can also assist Image-based Generative AI to generate AIGC (Artificial Intelligence Generated Content) using prompt engineering (i.e., users can drive the computers to provide the prompts for image generation purposely during the conversation with LMM). In this research, the authors designed multiple experiments to apply LMMs (i.e., Tencent YuanBao-interfaced DeepSeek and Hunyuan) to generate the prompts using prompt engineering. Based on the different prompt outputs, AIGC-based waste images were generated in Stable Diffusion. Correspondingly, generated images were trained in YOLOv8x models with the assistance of the annotations derived from previous trained AIGC waste model. Based on the needs of low-altitude waste monitoring, the authors tested the trained YOLOv8x models using benchmark dataset (LAD-RWB, Low-Altitude Drone Riparian Waste Benchmark). In conclusion, LMM-derived prompt-based AIGC have the possibility of applying in the low-altitude drone waste monitoring with over 0.8 precision (0.426 recall and 0.491 mAP50) in detecting the waste under the riparian background using a tiny test dataset (16 images). LMM can provide reasonable prompts for Stable Diffusion image generations.

Key Words : *Artificial Intelligence; Large Multimodal Model; Unmanned Aerial Vehicle; Waste; River*

1. INTRODUCTION

Until now, camera-based riparian environmental monitoring is always a challenging task that needs professional personnel to understand the on-site situation in a short time. Simultaneously, training the young labors to become professional personnel always needs time costs, which cannot solve the current tasks immediately. These years, camera platforms (smartphone^{1),2}, 4K camera^{3),4}) and drones⁵⁾⁻⁷⁾) have provided powerful supports for the professional personnel to understand the platform-surround situations. Nevertheless it still needs human eyes to observe and judge. Correspondingly, the new technologies are in considerate to make effort to improve the current situation.

Currently instead of human, Artificial Intelligence using computer vision (You Look Only Once, YOLO

series) can already deliver reasonable observation results¹⁾⁻⁷⁾. Correspondingly, data for training is still not enough under some specific scenes or backgrounds⁸⁾. Following, image generation model, i.e., Stable Diffusion (SD) in this research⁹⁾⁻¹⁰⁾, has not only shown its ability in the practical projects and also already increased the possibility of generating the images (i.e., AI-Generated Content, AIGC) for data augmentation, or in some cases, AIGC can solve the practical tasks solely. The reason is that AIGC can provide similar features as the practical world, which includes but not limited to color, shape and texture. Consistently, how to provide the necessary real-world-similar features in the generated images using Stable Diffusion, which is becoming more and more attractive theme to the researchers. At this moment, Large Multimodal Model (LMM) provided a solution that can drive image generation algorithm to provide the necessary real-world-similar features in the generated images. In recent years, LMMs have

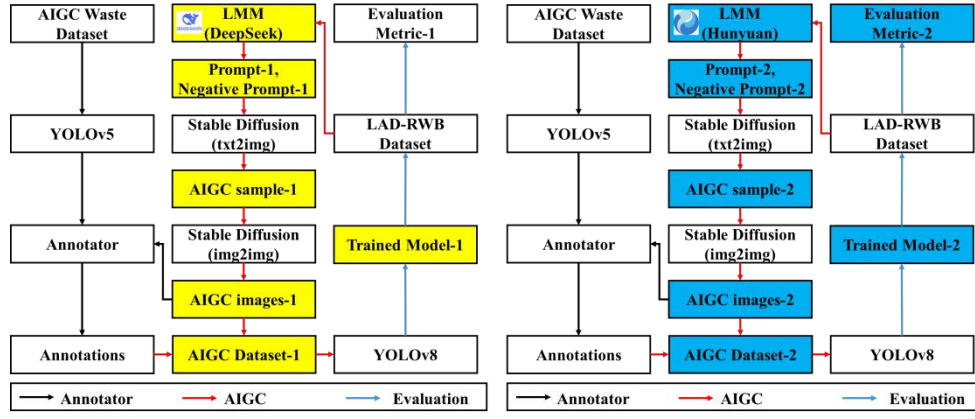


Fig.1 Workflow of applying the AIGC-based waste dataset in the low-altitude drone waste monitoring. There are 3 sections in the whole research: 1. Training one YOLOv5 model using AIGC Waste Dataset as the annotation generator; 2. Applying prompt engineering in LMMs to generate prompts for image generation, annotating the corresponding AIGC images using annotation generator and collecting the AIGC Dataset (images and annotations); 3. Using the AIGC Dataset to train the model and test using LAD-RWB Dataset for Evaluation Metric.

Table 1. Parameters setting

Parameters	Configuration (Stable Diffusion)	Configuration (YOLOv5)
Operating system (Version)	Windows (Win 11)	Linux (Ubuntu 20.04.4 LTS)
Model version	v1-5-pruned-emaonly.safetensors	v 6.0
GPU (VRAM)	NVIDIA GeForce RTX 3060 Laptop (6 GB)	NVIDIA GeForce RTX 3090 Ti (24 GB)
CUDA	11.8	11.7
Driver Version	522.06	515.65.01
Image-size (pixel)	608×608	1024×1024
Initial learning rate	-	0.01
Final learning rate	-	0.1
Optimizer	-	SDG
Momentum	-	0.937
Batch size	1	12
Batch count	100	-
Epochs	-	500
Patience	-	100

been applied in multiple scenes (i.e., Tencent¹¹⁾-interfaced DeepSeek¹²⁾ and Hunyuan¹³⁾ in this research¹⁴⁾⁻²⁰⁾. Based on the results, systematic prompts used in the conversations can help LMMs to understand the purpose and main content derived from the users clearly²¹⁾⁻²⁵⁾.

Considering of YOLO-, AIGC- and LMM-related potential in riparian environmental monitoring, the authors should develop the workflow including the details, i.e., how to apply these new technologies in the practical applications (i.e., waste detection) systematically. As depicted in **Fig.1**, there are 3 sections included as follows, DeepSeek and Hunyuan, individually:

- Section-1, training “AIGC Waste Dataset”⁶⁾ in YOLOv5 to become the annotation generator in this research and annotating the AIGC images in Section-2. AIGC Waste Dataset is a SD-derived synthesis dataset that includes four types of objects in the riparian environment, i.e., card board, bicycle, plastic bottle and plastic bag;

- Section-2, using LMM (DeepSeek R1 or Hunyuan T1) to generate prompts for Stable Diffusion image generation. During the process of the generation, one LAD-RWB (Low Altitude Drone Riparian Waste Benchmark Dataset, LAD-RWB⁸⁾ in the following content) sample selected from Section-3 should be used in the img2img (an image generation function in Stable Diffusion) as reference. And the authors collected the AIGC Dataset (images and annotations) ready for the model training in Section-3;
- Section-3, training AIGC Dataset in Section-2 and saving as Trained Model, then using LAD-RWB Dataset for Evaluation Metric on Trained Model, including P (Precision), R (Recall), mAP50 and mAP50-95.

2. METHODS

(1) YOLOv5 (Waste Dataset, Annotator)

As shown in **Table 1**, the authors showed the parameters settings of the trained YOLOv5 model using AIGC Waste Dataset for generating the annotations of AIGC Datasets. As shown in **Fig.2**, there are 5 steps in the flow chart of generating the AIGC images⁶⁾:

1. Selecting a Stable Diffusion checkpoint, i.e., v1-5-pruned-emaonly.safetensors [6ce0161689];
2. Using prompts (Objects, i.e., bikes, cardboard, PET bottles and plastic bags; Resolution, i.e., high resolution, 8k camera; View angle, i.e., bird view, UAV view; Area, i.e., riparian area) in txt2img;
3. Adjusting the Width/Height of image generations (i.e., 608/608 pixels);
4. Adjusting the Batch count/size of image generations (i.e., 100/1);
5. Starting the generation.

As shown in **Fig.3 & 4**, the authors showed results and label distribution statistics of Annotator. And correspondingly, as shown in **Table 2**, annotators should infer the AIGC images using the mentioned parameters for image sizes and confidence value.

(2) LMM

After preparing the trained model as the annotator, the authors should start to use LMM to understand and describe the provided image. As shown in **Fig.5**, it is a selected sample image from a dataset called “LAD-RWB: Low Altitude Dataset Riparian Waste Benchmark”. Noteworthy, image generation needs the information from different perspectives, including but not limited to color, texture, scene, and photography style. Next, as shown in **Table 3** (3-Steps AIGC Prompts Engineering Method or 3-Steps Method), the authors applied prompt engineering to drive the LMM to extract the necessary information from the sample image. 3-Steps Method include following 3 steps:

1. Understanding the image that the prompts will not overcome the scope;
2. Adjusting the prompt output and re-describing the image to bias the output;
3. Summarizing and categorizing the prompts for clear understanding.

With the assistance of 3-Steps Method, as shown in **Table 4**, LMMs extracted the necessary information in the shape of prompts from the provided sample image using DeepSeek R1 and Hunyuan T1, individually. **Table 4** (down-side) dictated the comparisons of prompts in different categories, which can be modified based on different purposes in the next step practical application.

(3) Stable Diffusion

As shown in **Fig.6**, the authors applied the prompts in fine-tuning Stable Diffusion WebUI online version `txt2img` function (<https://stablediffusion.com/>) to generate the reference images. Correspondingly, as shown in **Table 5 & 6**, the authors depicted the parameter setting for computer and Stable Diffusion WebUI local version `img2img` function, individually.

Fig.7 includes 2 grid images which visualized the comprehensive view of the generated images. Because of different LMMs, the photography types of the following grid images are different: DeepSeek R1 used murky water with more bottles than yellow bags in a cloudy weather, Hunyuan T1 used natural sunlight with balanced waste contributions. With the assistance of the annotator, all the AIGC images shown in **Fig.7** should be annotated using the mentioned annotator derived from AIGC Waste Dataset.

(4) YOLOv8x (DeepSeek R1)

Based on the purpose and costume, Fast-speed and balanced one-stage model YOLOv8x-det²⁶⁾ was chosen as the model for training the AIGC Datasets. 5 types of models (size changes from small to large, i.e., n, s, m, l and x) and 3 types of task modes (cls, classification; det, detection and seg, segmentation) can be chosen in this model. Following the parameters setting and image numbers shown in **Table 7 & 8**, the model was trained. **Fig.8 & 9** depicted the detailed process of training/validating and labels distribution statistics. **Fig.10** is a batch image sample of label (left-side) and corresponding prediction (right-side).

(5) YOLOv8x (HunYuan T1)

As same as DeepSeek R1-based Trained Model-1, Trained Model-2 followed the parameters setting and image numbers shown in **Table 9 & 10** to train YOLOv8x model. **Fig.11 & 12** depicted the detailed process of training/validating and labels distribution statistics. **Fig.13** also showed a batch image sample of label (left-side) and corresponding prediction (right-side). Tencent YuanBao is an AI platform that provides access to various models, including both DeepSeek and Hunyuan. We accessed both LMMs through the YuanBao interface for consistency.

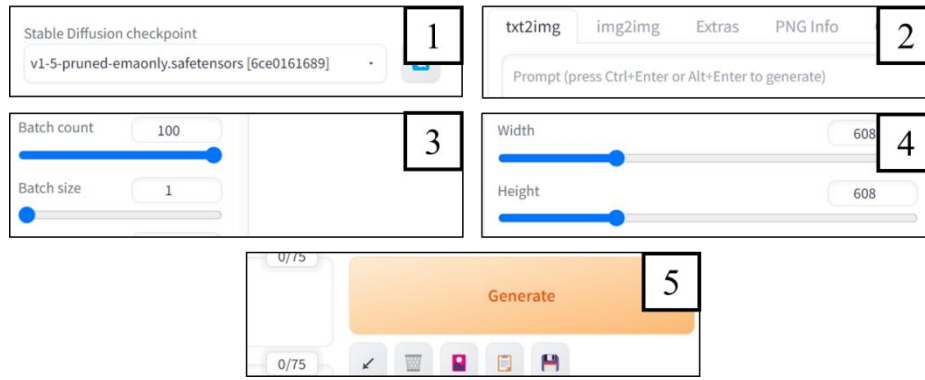


Fig.2 Flowchart of generating AIGC images in AIGC Waste Dataset.

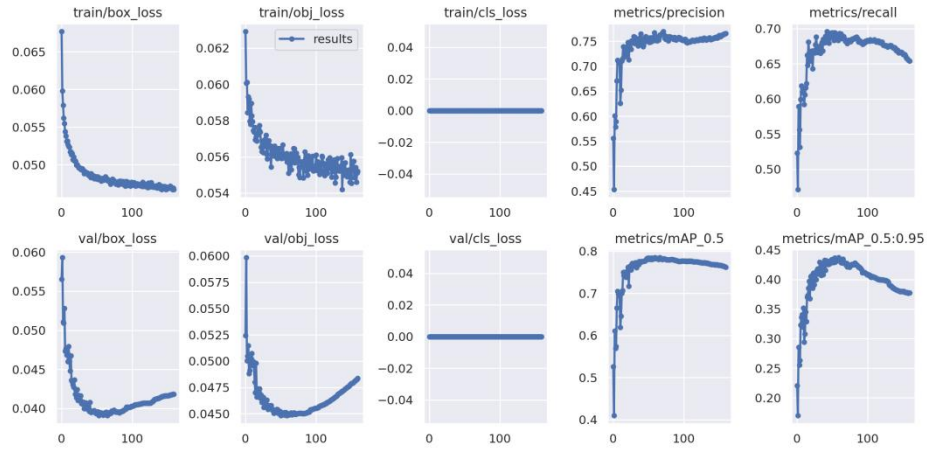


Fig.3 Results of training the model, which include train/valid-based box_loss/cls_loss/df_l_loss, precision/recall, and mAP50/50-95 (Annotator).

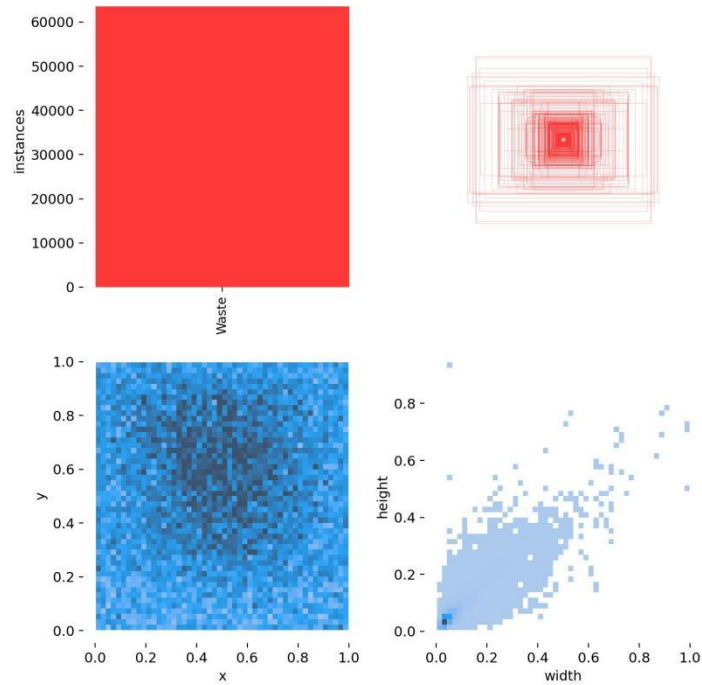


Fig.4 Label distribution statistics for training (Annotator, instances numbers, x-y, width-height and size of bounding box).

Table 2. Parameters setting (Annotation generations).

Specifications	Details
Imgsz (Image Size)	512 pixels
Conf (Confidence Value)	0.01, 0.1, 0.25

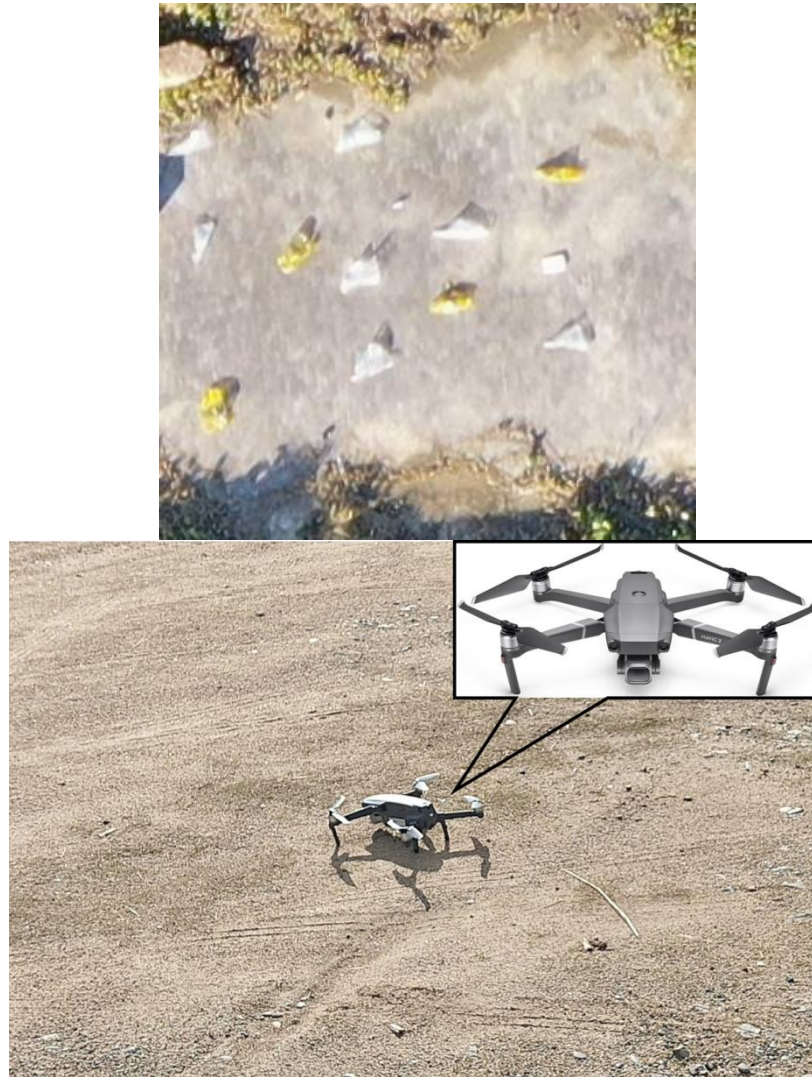


Fig.5 A random sample image (upper-side) chosen from LAD-RWB as the reference for prompt generations and on-site drone image (down-side, DJI Mavic 2 Air).

Table 3. 3-Steps AIGC Prompts Engineering Method using in the conversation with LMM.

Specifications	Details	Purposes
Step-1	Firstly, please explain what is in this image.	-Understanding the image
Step-2	In fact, it is a drone view waste detection image. This time, please describe this image again based on the information I provided.	-Adjusting the prompt output -Re-describing the image.
Step-3	I want to generate images based on the image I provided using Stable Diffusion, and I need the following factor included, i.e., Drone View, Waste Detection, Bottles, Transparent Plastic bags, Yellow Plastic bags, vegetation scenes, riparian environment. Please provide me prompts and negative prompts.	-Summarizing the prompts -Categorizing the prompts

Table 4. Prompts, Negative prompt and comparison in different categories (YuanBao-interfaced DeepSeek and Hunyuan, the prompts were crafted by the authors using YuanBao for iterative refinement).

Specifications		Details (DeepSeek R1)
Prompt-1		Masterpiece, best quality, 8K UHD, (drone aerial view:1.3), 30-100 meters height, top-down perspective, high-resolution environmental monitoring, (plastic bottles:1.4), apparent plastic bags, (yellow plastic bags:1.3), scattered debris accumulation, polluted riparian zone, muddy riverbank with sediment streaks, sparse reeds and grasses, wetland vegetation entanglement, murky shallow water, overcast lighting, ecological degradation documentation, depth of field, detailed texture, Beached Plastic Debris Index (BPDI)
Negative Prompt-1		EasyNegative, (worst quality, low resolution:1.3), human figures, vehicles, buildings, clean water, fish, trees, dense forest, sunset, rainbow, (text, signature, watermark:1.2), (intact containers, sealed bottles:1.1), cartoon, painting, 3D render, surrealism, floating plastic clusters

Specifications		Details (Hunyuan T1)
Prompt-2		"Aerial drone view of a riparian environment with a scattered waste detection scene: compacted dirt ground under soft natural sunlight, sparse wetland vegetation along a gentle creek bank, mid-air perspective showing multiple plastic bottles (clear/translucent, varied sizes) partially buried in soil, clusters of apparent plastic bags (white/semi-translucent, torn edges) and bright yellow plastic debris (strips, crumpled sheets) distributed unevenly across the frame, realistic texture details on litter and ground, muted earthy tones with high-contrast yellow highlights, 4K resolution, natural lighting with soft shadows from overhead foliage, no human figures or artificial structures, clean composition focusing on debris distribution."
Negative Prompt-2		"low resolution, blurry, cartoonish, overly saturated colors, crowded urban background, buildings, vehicles, people, animals, text, logos, glossy plastic (shiny reflections), vibrant green lawns, dense forests, mountains, snow, rain, night scenes, heavy shadows, perfect alignment of debris, metallic objects, glass, paper, organic waste, food scraps, medical waste, industrial debris, excessive clutter, abstract patterns, HDR effects, artificial lighting, underwater scenes, aerial shots with extreme tilt angles, cartoon filters, line art, pixelation, noise artifacts."

Specifications		Details (Comparison)
Structural Philosophy	DeepSeek R1	Using weighted token syntax (e.g., (item:1.3)), prioritizing algorithmic precision
	Hunyuan T1	Relying on descriptive language, prioritizing human-readable scene context.
Environmental Tone	DeepSeek R1	Stresses degradation (polluted, murky water, overcast).
	Hunyuan T1	Leaning toward neutral documentation (natural sunlight, muted tones).
Waste Specificity	DeepSeek R1	Quantifying debris importance (bottles > yellow bags).
	Hunyuan T1	Adding physical details: partially buried, torn edges, crumpled sheets.
Negative Prompt	DeepSeek R1	Blocking broad categories (e.g., EasyNegative, trees).
	Hunyuan T1	Surgically removing distractions (glossy plastic, perfect alignment).
Technical Intent	DeepSeek R1	BPDI suggests scientific indexing.
	Hunyuan T1	Clean composition implies ML training data optimization.
For Practical Use	DeepSeek R1	Generating imagery focused on pollution severity metrics.
	Hunyuan T1	Visually balanced, algorithm-friendly datasets without artifacts.
R1+T1		Using R1's weighted tokens + T1's debris physicality descriptions for optimal results.



Fig.6 2 generated waste image sample (1024 × 1024 pixels).

Table 5. Parameters setting (computer).

Specification	Details
OS	Ubuntu 22.04.4 LTS
CUDA Version	12.4
NVIDIA GPU Driver	550.144.03
GPU	NVIDIA GeForce RTX 3090

Table 6. Parameters setting (img2img).

Specifications	Details
SD checkpoints	v1-5-pruned-emaonly.safetensor [6ce0161689]
Resize mode	Crop and resize
Sampling steps	80
Sampling method	DPM++ 2M
Schedule type	Automatic
Width/Height	512/512 pixels
Batch count	100 (maximum, default=1)
Batch size	8 (maximum, default=1)
CFG Scale	7 (default)
Denoising strength	0.75 (default)

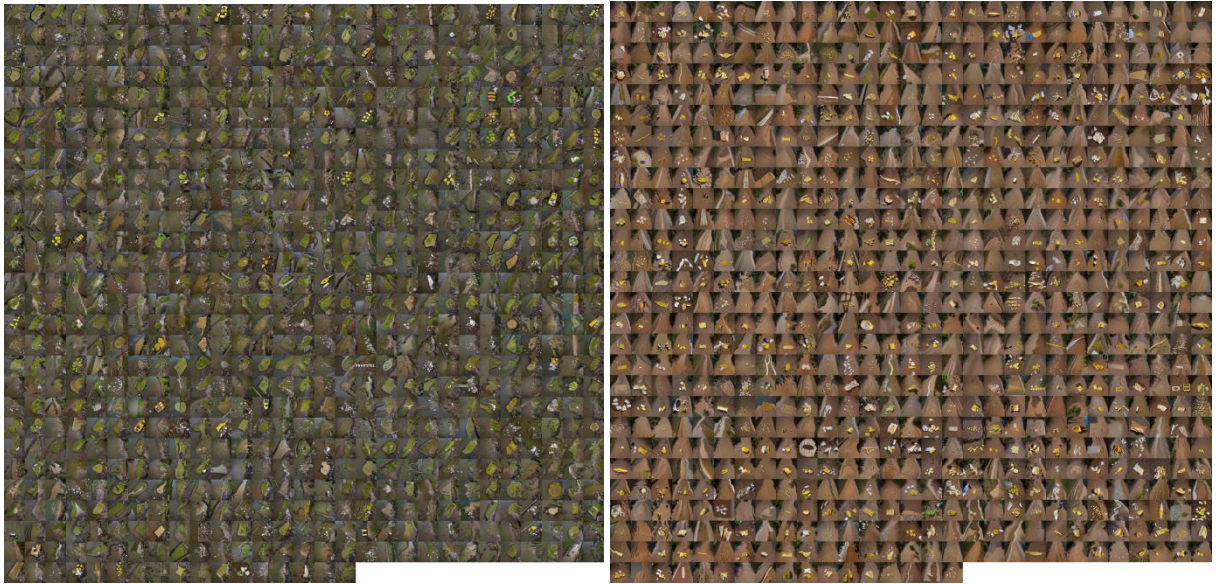


Fig.7 Grid images of all generated images (800 images, DeepSeek R1 and Hunyuan T1, img2img).

Table 7. Parameters setting (YOLOv8x, DeepSeek R1).

Specification	Details
Task	Detection
Model	yolov8x
Patience	100
Batch size	32
Imgsz	512 pixel
Epochs	500
Optimizer	Auto
Lr0	0.01
Momentum	0.937

Table 8. Image numbers (Trained Model-1, DeepSeek R1).

Specifications	Details
Train	600
Valid	200
Test	16

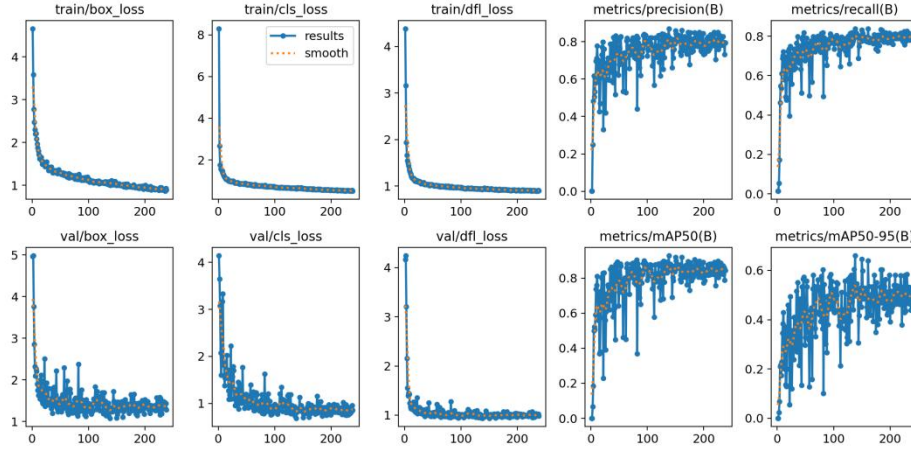


Fig.8 Results of training the model, which include train/valid-based box_loss/cls_loss/dfl_loss, precision/recall, and mAP50/50-95 (Trained Model-1, DeepSeek R1).

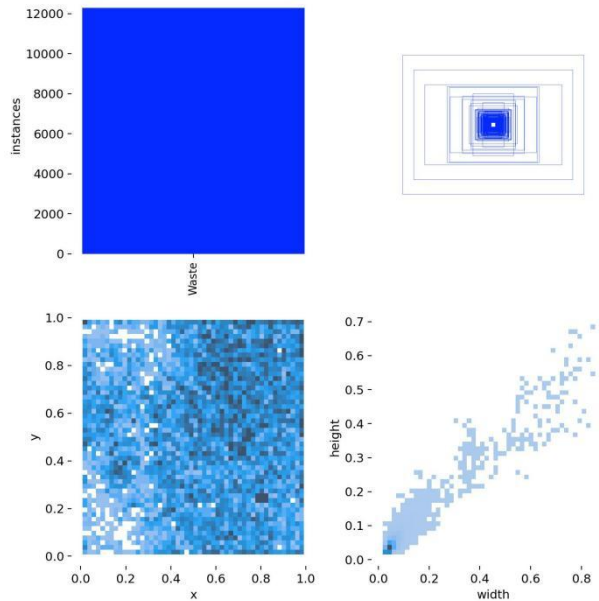


Fig.9 Label distribution statistics for training (Trained Model-1, instances numbers, x-y, width-height and size of bounding box).



Fig.10 Label and prediction samples of the batch images used for validation (Trained Model-1, DeepSeek R1)

Table 9. Parameters setting (YOLOv8x, Hunyuan T1).

Specification	Details
Task	Detection
Model	yolov8x
Patience	100
Batch size	32
Imgsz	512 pixel
Epochs	500
Optimizer	Auto
Lr0	0.01
Momentum	0.937

Table 10. Image numbers (Trained Model-2, Hunyuan T1).

Specifications	Details
Train	600
Valid	200
Test	16

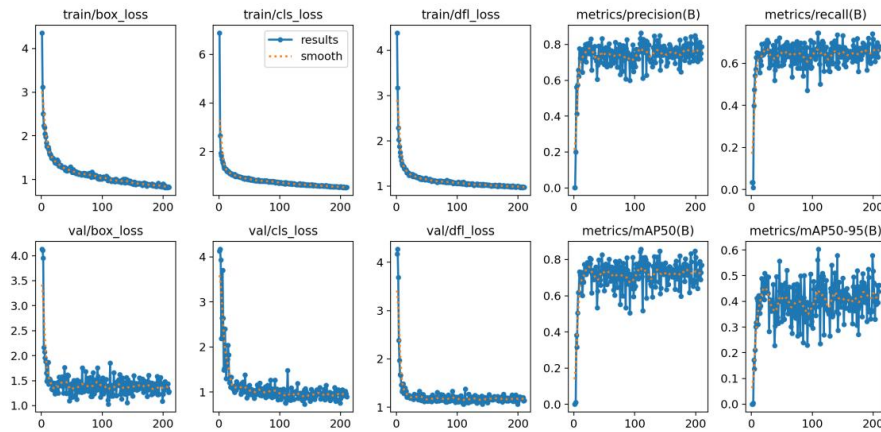


Fig.11 Results of training the model, which include train/valid-based box_loss/cls_loss/df_l_loss, precision/recall, and mAP50/50-95 (Trained Model-2, Hunyuan T1).

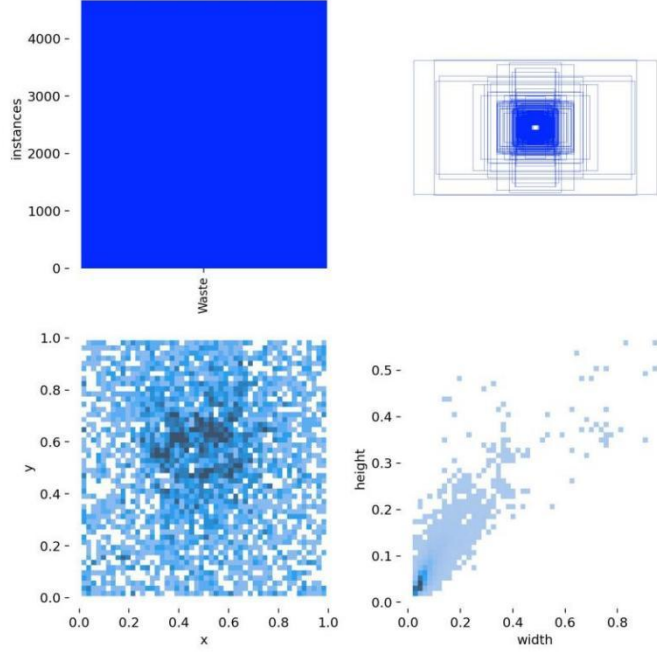


Fig.12 Label distribution statistics for training (Trained Model-2, instances numbers, x-y, width-height and size of bounding box).

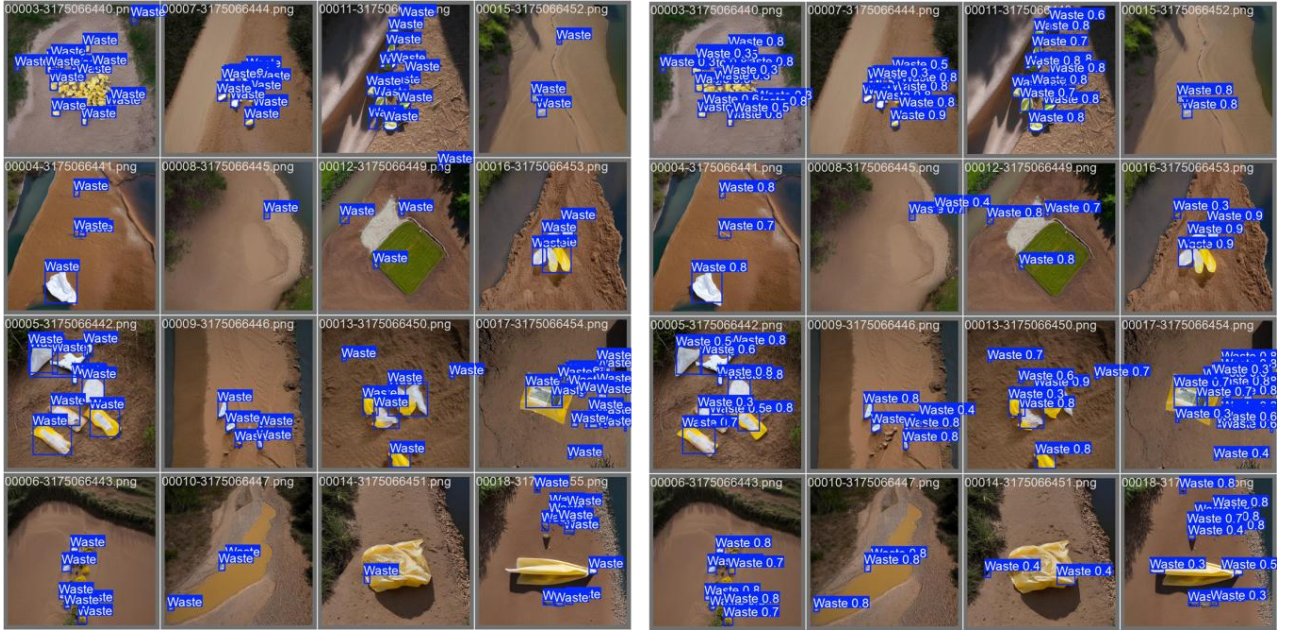


Fig.13 Label and prediction samples of the batch images used for validation (Trained Model-2, Hunyuan T1).

3. RESULTS

After collecting the mentioned 2 trained models, the authors tried to use parts of a benchmark dataset (16 images, 342×342 pixels, 100-m flight height) to evaluate them, individually. As shown in **Table 11 & 12**, the authors depicted the index used for the evaluations, i.e., Recall, Precision, mAP50 and mAP50-95 using 512 pixels image size and 0.001 confidence value. **Fig.14** showed the original, label and prediction samples of the batch images used for

validation, and **Table 13** compared the corresponding index between Trained Model-1 and -2 for the Evaluation Metric. Trained Model-1 derived from DeepSeek R1-generated prompts has advantages in all the evaluation index. Noteworthy, as shown in **Fig.15**, the detected waste objects derived from Trained Model-1 have much higher confidence value comparing with Trained Model-2 different objects (apparent plastic bag, plastic foams and yellow plastic bag in Sample-1, Sample-2 & -4 and Sample-3, individually) under different backgrounds.

Table 11. Confusion matrix.

		Predicted	
		Positive	Negative
True	Positive	True Positive (<i>TP</i>)	False Negative (<i>FN</i>)
	Negative	False Positive (<i>FP</i>)	True Negative (<i>TN</i>)

Table 12. Parameters setting (image size and confidence value for evaluation metric).

Specifications	Details
Imgsz (Image Size)	512 pixels
Conf (Confidence Value)	0.001

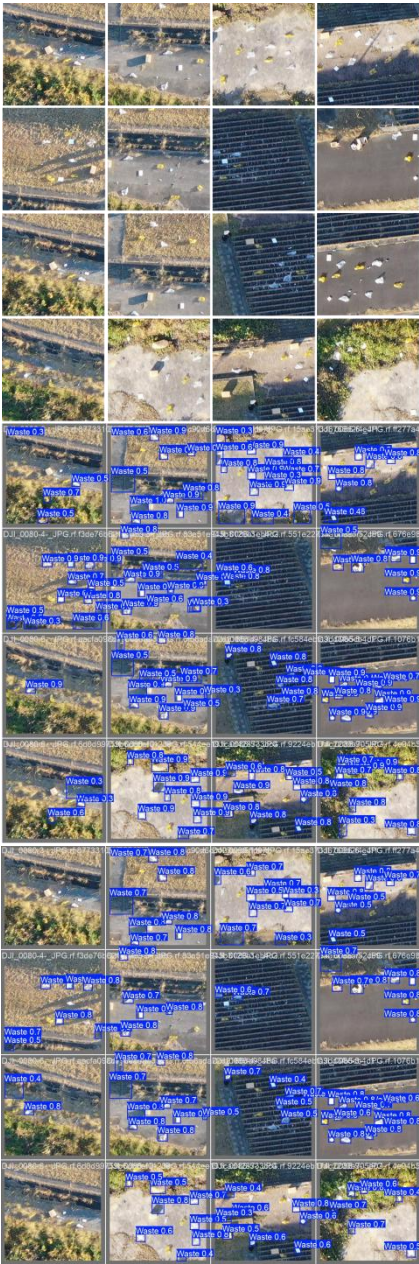


Fig.14 Original, label and prediction samples of the batch images used for validation.

Table 13. Evaluation metric based on different prompt-derived trained models

	P	R	mAP50	mAP50-95
Trained Model-1 (DeepSeek R1)	0.819	0.426	0.491	0.232
Trained Model-2 (Hunyuan T1)	0.67	0.409	0.43	0.204



Fig.15 Original and zoom-in prediction results of the batch images used for validation.

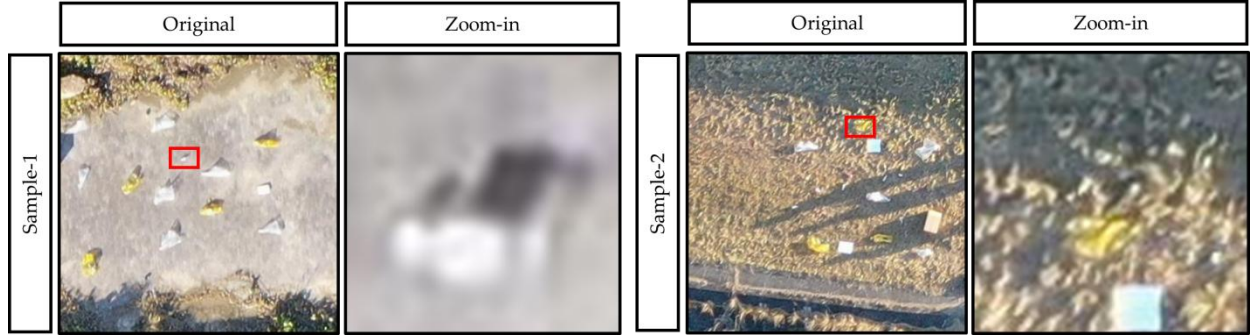


Fig.16 Original and zoom-in predict-impossible objects.

Table 14. Evaluation metric based on different image numbers using DeepSeek R1 prompts.

Prompts	Image Numbers	P	R	mAP50	mAP50-95
DeepSeek R1	800	0.819	0.426	0.491	0.232
	1600	0.782	0.448	0.516	0.235
	2400	0.733	0.511	0.613	0.309



Fig.17 Grid images of all generated images (800 images, DeepSeek R1, txt2img).

Fig.16 showed the objects that cannot be detected by both Trained Models, plastic bottle is too small for the detection and yellow plastic bag is not easy-to-detected between the background and its outline. Considering of the instances limitation derived from the image numbers, as shown in **Table 14**, the authors increased the images step by step (800 images each step). By increasing the image numbers, mAP50 has been changed from 0.491 to 0.613, i.e., to some degree, increasing the diversity and quantity of the AIGC Dataset can improve the detection accuracy.

4. DISCUSSION

In this research, the authors applied 3-Steps Method in LMMs to extract the necessary information from one specific image, and understanding the image systematically, which included but not limited to object, texture, color, scene and photography style. After summarizing and categorizing the mentioned prompts using prompt engineering, the extracted information can be used for image generation directly. And with the assistance of DeepSeek R1-based AIGC, Trained Model-1 showed its advantages comparing with Trained Model-2 (Hunyuan T1) in all cases of Evaluation Metric in the low-altitude drone-view waste monitoring.

Considering of the reasons, Trained Model-1 was trained using more instances than Trained Model-2 (around 12000 and 4000, individual), which is realistic. As shown in **Table 4**, the prompts derived from DeepSeek R1 are recommended to “generate imagery focused on pollution severity metrics”, which lead to the image generations of waste or litter pollutions in the riparian area. And Hunyuan T1 generated the prompts “mid-air perspective showing multiple plastic bottles (clear/translucent, varied sizes) partially buried in soil” and “realistic texture details on litter and ground” for much clearer individual objects- and background-related descriptions. Based on the differences between the prompts, Trained Model-1 derived from DeepSeek R1 is much appropriate to the benchmark dataset (16 images, 342×342 pixels, 100-m flight height) in this research.

Noteworthy, the authors also tried local txt2img function in Stable Diffusion directly to generate the images. As shown in **Fig.17**, although riparian-related features have been generated correctly, waste-based objects have not been generated as same as img2img function, even the waste-objects-related prompts have already been

included. Worth mentioning, if the reference images in img2img have already been generated using fine-tuned model, the image generations are more controllable. Because the existing reference images can be judged and adjusted firstly, with the assistance of reference images, img2img function is more stable than txt2img in the image generation tasks (which was mainly based on the human judgment).

The systematic novel pipeline (LMM $\rightarrow$ AIGC $\rightarrow$ Annotator $\rightarrow$ Detector) demonstrates a path toward generating large-scale, task-specific synthetic training data with minimal human intervention, starting from just a single reference image and a conversational interface with an LMM. This approach is particularly valuable for applications where collecting and manually annotating thousands of real-world images is prohibitively expensive or logistically challenging.

5. CONCLUSION

LMM-derived prompt-based AIGC have the possibility of applying in the low-altitude drone waste monitoring (LAD-RWB-based) with over 0.8 precision in detecting the waste under the riparian background. After increasing the image numbers, mAP50 has been improved from under 0.5 to over 0.6. LMM can provide reasonable prompts for Stable Diffusion image generations using novel 3-Steps Method. YOLOv5 “annotator” was derived from real-world dataset called Nisha484, which can be considerate as a trustable annotator. And in conclusion, to some degree, codifying systematic biases and false boxes was ignored. But when generating the synthetic data, the generated content was limited in the specific objects, in the authors’ opinion, which can be accepted. And in the previous works, comparing with the real-world general dataset, specific-designed synthetic data annotated by the trained annotator had much higher accuracy in the specific waste detection metric. And the automatic annotator should be used or considerate as a speed-up rough annotation provider for any types of objects. After the rough annotation, manual corrections are necessary for the higher accuracy. Rather than the higher accuracy, faster speed for annotating the real-world data with controllable errors is our target.

6. LIMITATIONS

Even the method of using LMM as the prompt generator for Stable Diffusion, has been proved its possibility, the limitation of the dataset amount for test still exists in this research. Because all the AIGC data for training were generated based on the test data sample, in this research the authors only consider the limited test dataset. The small test set size is a major limitation that affects the statistical reliability and generalizability of our findings, the results in this research are indicative rather than conclusive. This research is basically discussing the possibility of applying the LMM to generate the necessary data using prompt engineering, and it is a proof of concept instead of practical application (or not yet).

7. FUTURE WORK

The method applied in this research focused on the waste monitoring task observed from close-distance drone-view. The prompt-engineering derived from researchers can applied in the Edge AI devices⁽²⁷⁻³¹⁾ to generate dataset including both images and annotations automatically. And the application of edge AI devices can support the indoor operators to reduce the workload. And researchers can connect prompt engineering with Large Language Models⁽³²⁻³⁸⁾, Text-To-Speech⁽³⁹⁻⁴⁷⁾ and Automatic-Speech-Recognition⁽⁴⁸⁻⁵¹⁾ together in the drone-platform Edge AI devices without any other unnecessary operation just using human voice. The method in this research was designed for low-altitude drone-view waste monitoring. Prompts should be accumulated based on the 3-Steps Method and applied under more practical engineering scenes, particularly drone-platformed applications. Excepting the mentioned content, there are also several bulleted points that can deepen the research discovery, as follows:

- Waste Specificity-related Prompts;
- Environmental Tone-related Prompts;
- Structural Philosophy-related Prompts;
- Scenes-related Prompts;
- Technical Intent-related Prompts;
- Negative Prompts;

We will discuss specific factors contributing to the gap, such as the lack of complex occlusions, unrealistic textures, lighting inconsistencies in AIGC images, and the inability to generate very small or heavily degraded objects (as seen in Fig. 16).

We will discuss potential mitigation strategies, including domain adaptation techniques, using more diverse reference images for img2img, incorporating negative prompts more effectively to avoid

unrealistic features, and the potential of using real background images with synthetic waste overlays.

We will provide a quantitative analysis of the failure cases, for instance, by reporting the average size (in pixels) of missed objects compared to detected ones, and discussing the impact of IoU thresholds on the low recall.

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